

1. Suppose A is an $n \times n$ orthogonal matrix, and $\mathbf{x} \in \mathbb{R}^n$ prove or disprove that $\|A\mathbf{x}\| = \|\mathbf{x}\|$, where $\|\bullet\|$ is the length of the vector. No partial marks will be given if you take a specific numerical problem and solve.

Solution:

$$\|A\mathbf{x}\|^2 = (A\mathbf{x}) \bullet (A\mathbf{x})$$

$$\Rightarrow = (A\mathbf{x})^T (A\mathbf{x})$$

$$\Rightarrow = \mathbf{x}^T A^T A \mathbf{x}$$

$$\Rightarrow = \mathbf{x}^T \mathbf{x} = \|\mathbf{x}\|^2 \quad (A^T A) = I \text{ as } A \text{ is an orthogonal matrix.}$$

$$\text{Thus, } \|A\mathbf{x}\| = \|\mathbf{x}\|$$

2. Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors in $\mathbb{R}^2$, such that $A = \mathbf{u}\mathbf{v}^T$. Obtain the SVD of the matrix A . No partial marks will be given if you take a specific numerical problem and solve.

Solution: Given that $A = \mathbf{u}\mathbf{v}^T$, the matrix A is a 2×2 matrix with rank 1. This is because the outer product of two vectors will result in a rank 1 matrix. Thus, the matrix A is non-invertible. Also assume that the vectors $\mathbf{u}$ and $\mathbf{v}$ are unit vectors.

$$A^T A = (\mathbf{u}\mathbf{v}^T)^T (\mathbf{u}\mathbf{v}^T) = \mathbf{v}\mathbf{u}^T \mathbf{u}\mathbf{v}^T = \mathbf{v}\mathbf{v}^T.$$

Also, $AA^T = (\mathbf{u}\mathbf{v}^T)(\mathbf{u}\mathbf{v}^T)^T = \mathbf{u}\mathbf{v}^T \mathbf{v}\mathbf{u}^T = \mathbf{u}\mathbf{u}^T$. Let us now find the eigenvalues and eigenvectors of $A^T A$ and AA^T .

$A^T A$

AA^T

$$A^T A = \mathbf{v}\mathbf{v}^T$$

$$\Rightarrow A^T A \mathbf{v} = \mathbf{v}\mathbf{v}^T \mathbf{v}$$

$$\Rightarrow A^T A \mathbf{v} = (\mathbf{v}^T \mathbf{v}) \mathbf{v}$$

$$\Rightarrow A^T A \mathbf{v} = 1 \mathbf{v} \quad (\because \mathbf{v}^T \mathbf{v} = 1)$$

$$AA^T = \mathbf{u}\mathbf{u}^T$$

$$AA^T \mathbf{u} = \mathbf{u}\mathbf{u}^T \mathbf{u}$$

$$\Rightarrow AA^T \mathbf{u} = (\mathbf{u}^T \mathbf{u}) \mathbf{u}$$

$$\Rightarrow AA^T \mathbf{u} = 1 \mathbf{u} \quad (\because \mathbf{u}^T \mathbf{u} = 1)$$

Thus the eigenvalue of $A^T A$ is 1 and the corresponding eigenvector is $\mathbf{v}$.

Thus the eigenvalue of AA^T is 1 and the corresponding eigenvector is $\mathbf{u}$.

The non zero eigenvalues of both the matrices are 1. So the non-zero singular values turn out to be 1. (If you did not consider the vectors to be unit vectors, then you gotta divide by the length of the two vectors. In this case, the singular values turn out to be the product of the lengths of the two vectors.

We know that the SVD of a matrix $A = U\Sigma V^T$, where U is the orthogonal matrix of eigenvectors of AA^T , Σ is the matrix of singular values of A and V is the orthogonal matrix of eigenvectors of $A^T A$. In our case, U is a 2×2 matrix, Σ is a 2×2 matrix and V^T is a 2×2 matrix.

Note that $A^T A$ and AA^T are symmetric matrices with distinct eigenvalues 1 and 0. Therefore, the eigenvectors are orthogonal to each other. Hence the eigenvectors corresponding to the eigenvalue = 0 are $\begin{pmatrix} -u_2 \\ u_1 \end{pmatrix}$ for AA^T and $\begin{pmatrix} -v_2 \\ v_1 \end{pmatrix}$ for $A^T A$ respectively. Therefore, we have

$$A = U\Sigma V^T = \begin{pmatrix} u_1 & -u_2 \\ u_2 & u_1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 & v_2 \\ -v_1 & v_2 \end{pmatrix}.$$

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- Given a singular value decomposition of an $m \times n$ matrix $A = U \Sigma V^T$, find an SVD of A^T . How are the singular values of A^T related to A ?

Solution:

$$\begin{aligned} A &= U\Sigma V^T \\ \Rightarrow A^T &= (U\Sigma V^T)^T \\ \Rightarrow A^T &= (V\Sigma^T U^T)^T \end{aligned}$$

We also see that the singular values of A and A^T are the same.

- Show that if $\mathbf{x}$ is in both W and $W^\perp$, then $\mathbf{x} = \mathbf{0}$. No partial marks will be given if you take a specific numerical problem and solve.

Solution: Let $\mathbf{x}$ be a vector in $W \cap W^\perp$. Since $\mathbf{x} \in W$, it is perpendicular to all the vectors in $W^\perp$. Also, since $\mathbf{x} \in W^\perp$, $\mathbf{x}$ is perpendicular to all the vectors in W . Recall that W and $W^\perp$ are orthogonal complement subspaces and that we know that the bases for W and $W^\perp$ together span the larger vector space V .

This means the vector $\mathbf{x}$ is orthogonal to all the vectors in V and since it is also an element of V , it is orthogonal to itself. The only vector orthogonal to all the vectors and to itself is the zero vector. Hence $\mathbf{x} = \mathbf{0}$.

5. If P is a projection matrix, prove or disprove $(I - P)$ is also a projection matrix, where I is the identity matrix. Hint: Recall the definition of Projection matrix.

Solution: Let us first understand what $(I - P)$ matrix is. Let us look at $(I - P)\mathbf{x}$, where $\mathbf{x} \in \mathbb{R}^n$.

$(I - P)\mathbf{x} = I\mathbf{x} - P\mathbf{x} = \mathbf{y}$ where $\mathbf{y}$ is the perpendicular component of $\mathbf{x}$. This component is nothing but the projection of the vector $\mathbf{x}$.

$$(I - P)^T = (I^T - P^T) = I - P \quad (I - P) \text{ is therefore symmetric}$$

$$(I - P)^2 = (I - P)(I - P) = I^2 - 2IP + P^2 = I - 2P + P = I - P$$

Thus $(I - P)$ is both symmetric and idempotent. Hence $(I - P)$ is a projection matrix.

6. Let $M = I - 2\mathbf{u}\mathbf{u}^T$ where $\mathbf{u}$ is a unit vector in $\mathbb{R}^2$. This matrix M reflects every vector about the line passing through the origin and along the direction of $\mathbf{u}$. Obtain the eigenvalues of the matrix M .

Solution: Let $M\mathbf{x} = \lambda\mathbf{x}$ We know that applying reflection transformation once, the left appears right and right appears left. If we apply the reflection second time, we get left appearing left and right appearing right.

Thus

$$\begin{aligned}M(M\mathbf{x}) &= M(\lambda\mathbf{x}) = \lambda(M\mathbf{x}) = \lambda^2\mathbf{x} \\ \Rightarrow M(M\mathbf{x}) &= M^2\mathbf{x} = I\mathbf{x} = \lambda^2\mathbf{x} \\ &\Rightarrow \mathbf{x} = \lambda^2\mathbf{x} \\ \Rightarrow (\lambda^2 - 1)\mathbf{x} &= \mathbf{0} \\ \lambda &= \pm 1\end{aligned}$$

Thus the eigenvalues of any reflection matrix are ± 1 .

7. Prove that for a real symmetric matrix A , the eigenvectors corresponding to distinct eigenvalues are orthogonal to each other.

Obtain the conditions for solution to exist for the system of linear equations $A\mathbf{x} = \mathbf{b}$. Assume A is a tall matrix with rank $r < n$, where n is the number of columns of A .

Solution: Refer the notes you had written in the class!